

Test Case Design Assignment

Intelligent AI-Based Smart Crop Disease Diagnosis and Yield Prediction System CSA1017-SOFTWARE ENGINEERING

Name: - G. Ram Charan
Reg no: -192525043

1. Problem Statement Overview

1.1 Industry Context

Agriculture is increasingly adopting AI technologies to improve crop productivity and minimize losses caused by diseases and pests. Computer vision, drones, IoT sensors, and weather analytics help farmers monitor crop conditions and implement precision farming practices. The system under test brings these capabilities together into a single, farmer-facing platform.

1.2 Problem Statement

Crop diseases significantly reduce agricultural productivity and often spread before farmers can detect them through manual inspection. Existing methods are time-consuming and require expert intervention. The Intelligent AI-Based Smart Crop Disease Diagnosis and Yield Prediction System analyzes crop images, monitors environmental conditions, predicts disease outbreaks, estimates crop yield, and recommends suitable treatments, in order to support sustainable agriculture while improving farm productivity and profitability.

1.3 Objectives

The system aims to detect crop diseases using AI, predict crop yield accurately, monitor environmental conditions continuously, recommend suitable disease treatments, analyze weather impacts on crop health, generate farm analytics reports, reduce overall crop losses, and improve agricultural productivity for both smallholder farmers and enterprise farm operators.

1.4 Expected Outcomes

Successful implementation is expected to result in early disease identification, increased crop yield, reduced pesticide usage through targeted treatment, better and faster farming decisions, improved farm profitability, more sustainable agricultural practices, enhanced food security, and overall more efficient crop management.

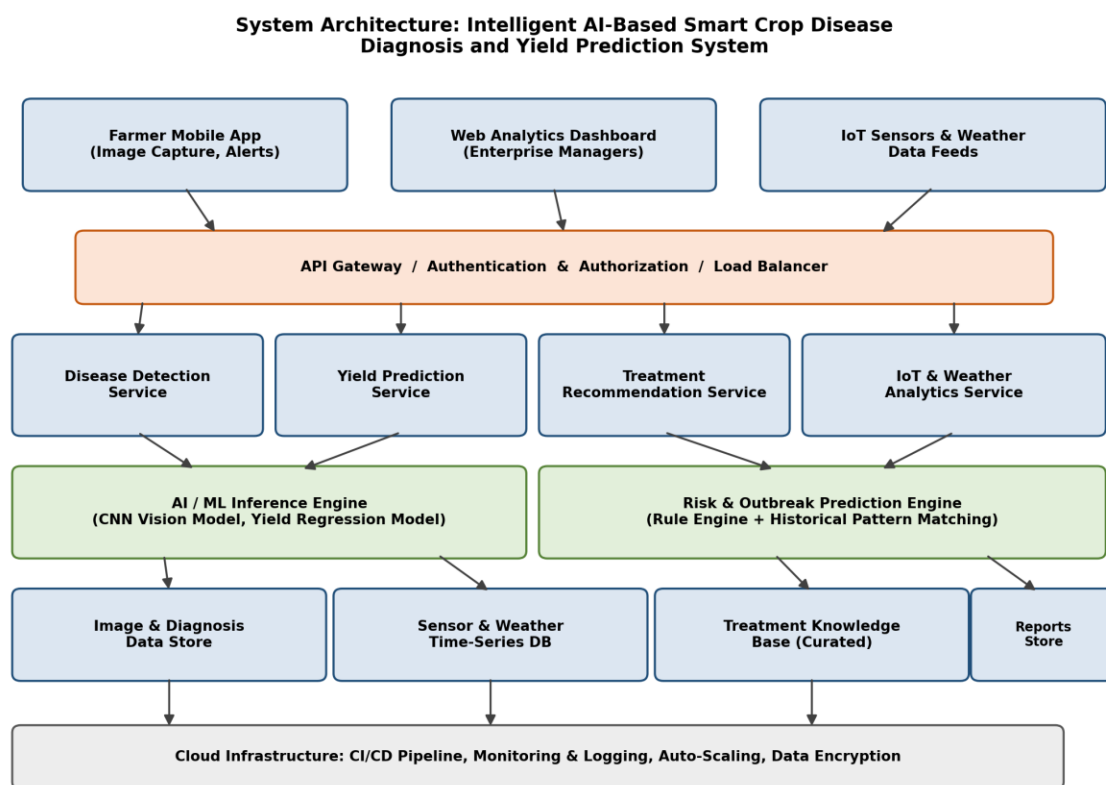
1.5 Purpose of This Document

This document defines the test case design for the assigned problem statement. It identifies the functional and non-functional requirements to be tested, defines the relevant test

scenarios, documents the system architecture that the test cases are derived from, applies recognized test design techniques, and presents a complete, traceable set of test cases covering normal, boundary, invalid, and exceptional conditions.

2. System Architecture

The architecture below represents the logical structure of the system under test. Test scenarios and test cases in this document are mapped against each layer of this architecture to ensure that requirement and functional coverage extends across the client applications, the service layer, the AI/ML core, the data layer, and the underlying cloud infrastructure.



2.1 Architecture Layers

- Client Layer: Farmer mobile application for image capture and alerts, a web analytics dashboard for enterprise farm managers, and inbound IoT sensor / weather data feeds.
- Gateway Layer: A single API gateway handles authentication, authorization, request routing, and load balancing across all backend services.

- Service Layer: Independent services for disease detection, yield prediction, treatment recommendation, and IoT/weather analytics, allowing each capability to be tested and scaled independently.
- AI/ML Core: A CNN-based vision model and yield regression model perform inference, while a separate risk and outbreak prediction engine evaluates environmental readings against historical outbreak patterns.
- Data Layer: Dedicated stores for diagnosis images, sensor/weather time-series data, the curated treatment knowledge base, and generated reports.
- Infrastructure Layer: Cloud infrastructure providing CI/CD, monitoring and logging, auto-scaling, and encryption of data at rest and in transit.

This layered view is used throughout the document: functional requirements are grouped by service, and non-functional requirements are validated primarily at the gateway and infrastructure layers.

3. Functional Requirements (To Be Tested)

The following functional requirements were derived from the problem statement, objectives, and expected outcomes, and form the basis for the test scenarios and test cases in Sections 5 and 7.

- FR1. The system shall allow a farmer to capture or upload a crop leaf image and receive an automated disease diagnosis with a confidence score.
- FR2. The system shall highlight the infected region of a diagnosed leaf image using a boundary overlay.
- FR3. The system shall flag low-confidence diagnoses for manual review by an agronomist.
- FR4. The system shall allow registration and ingestion of IoT soil-moisture, temperature, and humidity sensor data.
- FR5. The system shall continuously evaluate environmental readings and issue proactive disease-outbreak risk alerts.
- FR6. The system shall accept crop type, acreage, and historical yield data and return an AI-generated yield forecast.

- FR7. The system shall retrieve and display organic and chemical treatment recommendations, including dosage and safety precautions, following a positive diagnosis.
- FR8. The system shall analyze weather impact on crop health using integrated weather APIs.
- FR9. The system shall generate and export monthly farm analytics reports in PDF format.
- FR10. The system shall provide a web dashboard for enterprise farm managers to monitor regional productivity and resource allocation.
- FR11. The system shall authenticate users and restrict access to authorized farmers, agronomists, and managers only.

4. Non-Functional Requirements (To Be Tested)

Non-functional requirements define the quality attributes that the system must satisfy in addition to its functional behaviour.

- NFR1. Performance: Disease diagnosis inference shall complete within 3 seconds for at least 95% of requests under normal load.
- NFR2. Scalability: The platform shall auto-scale to support at least 1,000 concurrent dashboard users and 500 concurrent image submissions without SLA breach.
- NFR3. Reliability: The platform shall maintain at least 99.5% uptime measured over a rolling 30-day period.
- NFR4. Security: All data in transit and at rest shall be encrypted, and accounts shall be locked after 5 consecutive failed login attempts.
- NFR5. Usability: The mobile application shall remain fully operable on entry-level Android devices with 2GB RAM.
- NFR6. Compatibility: The web dashboard shall render correctly across the latest two major versions of Chrome, Edge, and Safari.
- NFR7. Data Integrity: Sensor readings outside physically plausible ranges shall be discarded and flagged, not used in risk calculations.

5. Test Scenarios

Test scenarios were identified by walking through each functional requirement and each layer of the system architecture, covering normal usage, boundary conditions, invalid input, and exceptional/failure conditions.

5.1 Disease Detection Module

- Verify accurate diagnosis for clear, well-lit crop leaf images.
- Verify correct handling of healthy (disease-free) leaf images.
- Verify low-confidence results are routed to expert review.
- Verify rejection of unsupported file types and oversized images.
- Verify graceful recovery when the network drops mid-upload or the inference service is unavailable.

5.2 IoT & Weather Monitoring Module

- Verify sensor registration, live readings, and historical trend charts.
- Verify outbreak-risk alerts trigger correctly at and around defined humidity/temperature thresholds.
- Verify handling of faulty sensors, offline sensors, and unreachable weather APIs.

5.3 Yield Prediction Module

- Verify yield forecasts are generated for valid crop, acreage, and historical inputs.
- Verify acreage boundary handling at the supported minimum and maximum.
- Verify validation of missing, negative, or non-numeric input.
- Verify behaviour when historical data for a region is unavailable.

5.4 Treatment Recommendation Module

- Verify correct organic and chemical treatment recommendations are returned after diagnosis.
- Verify dosage details and safety precautions are displayed accurately.
- Verify handling of diseases not present in the treatment knowledge base.

- Verify recommendations remain available if the knowledge base is temporarily unreachable.

5.5 Analytics Dashboard & Reporting Module

- Verify PDF report generation and export for a selected period.
- Verify dashboard filtering by region and crop type.
- Verify handling of empty datasets and invalid (future) reporting periods.

5.6 Security & Non-Functional Scenarios

- Verify authentication, authorization, and account lockout behaviour.
- Verify encrypted data transmission for login and image upload.
- Verify system performance, scalability, and availability targets are met.

6. Test Design Techniques Applied

A combination of recognized test design techniques was applied to ensure that the test suite achieves both breadth and depth of coverage while keeping the number of test cases manageable.

6.1 Equivalence Partitioning

Input fields such as image file size, acreage, and confidence score were divided into valid and invalid partitions (for example, 'file size within limit' versus 'file size exceeds limit'), with one representative test case selected per partition. This is applied in test cases such as TC-DD-01, TC-DD-09, and TC-YP-06.

6.2 Boundary Value Analysis

For every numeric or threshold-driven condition, values were tested exactly at, just above, and just below the boundary. This technique is used extensively for the outbreak-alert humidity threshold (TC-IOT-04, TC-IOT-05), the risk temperature band (TC-IOT-06, TC-IOT-07), acreage limits (TC-YP-03 to TC-YP-05), the file-size limit (TC-DD-05, TC-DD-06), and the confidence-score thresholds (TC-DD-07, TC-DD-08, TC-TR-04).

6.3 Decision Table Testing

The outbreak-alert logic depends on a combination of conditions (humidity level, sustained duration, and temperature band). A decision-table approach was used to enumerate the meaningful combinations of these conditions and confirm that an alert is raised only when all qualifying conditions are true simultaneously, as exercised by TC-IOT-04 through TC-IOT-07 and TC-IOT-12.

6.4 State Transition Testing

The diagnosis workflow moves through defined states: Submitted, Processing, Diagnosed (Auto-Accepted), Flagged for Review, and Treatment Displayed. State transition testing was used to verify valid transitions (for example, Processing to Diagnosed) and to confirm that invalid transitions, such as reaching the Treatment screen without a completed diagnosis, are blocked, as exercised by TC-TR-06.

6.5 Error Guessing

Based on experience with similar upload-and-inference systems, additional exceptional scenarios were added for conditions not obviously implied by the requirements alone, including corrupted files, service outages, session timeouts, and sensor anomalies (TC-DD-11 through TC-DD-14, TC-IOT-09 through TC-IOT-11, TC-AN-05).

7. Detailed Test Cases

The table below is the single consolidated test-case table for this document. It contains the Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status for every test case identified in Section 5, grouped by module and covering normal, boundary, invalid, and exceptional conditions as required by the assignment instructions.

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-DD-01	Upload a clear, in-focus tomato leaf image for disease detection	1. Open mobile app 2. Login 3. Tap 'Diagnose Crop' 4. Capture/upload leaf image 5. Submit for analysis	JPEG image of tomato leaf with Early Blight symptoms, 2MB	System returns diagnosis 'Early Blight' with confidence score and highlighted infected region within 3 seconds	As expected	Pass
TC-DD-02	Upload a healthy leaf image with no visible disease	1. Login 2. Upload leaf image 3. Submit	JPEG image of healthy tomato leaf, 1.5MB	System classifies image as 'Healthy - No Disease Detected'	As expected	Pass
TC-DD-03	View disease history log after multiple diagnoses	1. Login 2. Navigate to 'History' tab 3. View past diagnoses	3 previously diagnosed images in account	All past diagnoses listed with date, disease name and confidence score	As expected	Pass
TC-DD-04	Low-confidence diagnosis flagged for expert review	1. Upload ambiguous/blurred leaf image 2. Submit for analysis	Leaf image with confidence score below 60%	Result flagged as 'Needs Expert Verification' and routed to agronomist queue	As expected	Pass
TC-DD-05	Upload image exactly at maximum allowed file size	1. Select image of exactly 10MB 2. Upload	Image file = 10 MB (upper boundary)	Upload succeeds and image is processed normally	As expected	Pass
TC-DD-06	Upload image just above maximum allowed file size	1. Select image of 10.1MB 2. Attempt upload	Image file = 10.1 MB	System rejects upload with message 'File exceeds maximum size of 10MB'	As expected	Pass
TC-DD-07	Confidence score boundary at expert-review threshold (60%)	1. Upload image producing exactly 60% confidence	Simulated model output = 60% confidence	Result is treated as auto-accepted (boundary is inclusive) as per business rule	As expected	Pass
TC-DD-08	Confidence score boundary just below threshold (59%)	1. Upload image producing 59% confidence	Simulated model output = 59% confidence	Result flagged for expert verification	As expected	Pass
TC-DD-09	Upload an unsupported file format	1. Select a .pdf file 2. Attempt upload as crop image	Sample .pdf document	System rejects file with error 'Unsupported file format. Use JPEG/PNG'	As expected	Pass

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-DD-10	Submit diagnosis request with no image selected	1. Tap 'Diagnose Crop' 2. Tap Submit without selecting an image	No file attached	System displays validation message 'Please select an image before submitting'	As expected	Pass
TC-DD-11	Upload a corrupted image file	1. Select a corrupted .jpg file 2. Upload	Corrupted JPEG (truncated header)	System detects invalid image and shows 'Unable to process image, please retry'	As expected	Pass
TC-DD-12	Network disconnects during image upload	1. Start upload 2. Disable network mid-transfer	Stable Wi-Fi image upload in progress	App detects failure, shows retry prompt, and preserves the queued image locally	As expected	Pass
TC-DD-13	AI inference service is temporarily unavailable	1. Submit image while backend ML service is down	Valid leaf image, ML service returns 503	App shows 'Diagnosis service unavailable, please try again shortly' and queues request	As expected	Pass
TC-DD-14	Concurrent submissions from many farmers overload inference queue	1. Simulate 500 concurrent image submissions	500 simultaneous requests from load-test harness	System auto-scales inference pods and processes all requests within SLA of 3 seconds each	As expected	Pass
TC-IOT-01	Connect a new soil-moisture sensor to the platform	1. Open 'Add Device' 2. Enter sensor ID 3. Confirm pairing	Valid sensor ID SM-1042	Sensor is registered and begins streaming readings every 5 minutes	As expected	Pass
TC-IOT-02	View real-time temperature and humidity dashboard	1. Login as Farm Manager 2. Open 'Environment' tab	Live feed from 3 registered sensors	Dashboard displays current readings updated within the last 5 minutes	As expected	Pass
TC-IOT-03	Historical sensor trend chart for the past 7 days	1. Open 'Environment' tab 2. Select 7-day range	7 days of stored sensor readings	Line chart renders temperature and humidity trends for the selected range	As expected	Pass
TC-IOT-04	Humidity exactly at outbreak-alert threshold (85%) for 48 hours	1. Simulate sensor humidity = 85% sustained for 48 hours	Humidity = 85%, Duration = 48h (boundary)	Outbreak risk alert is triggered (threshold inclusive)	As expected	Pass
TC-IOT-05	Humidity just below outbreak threshold (84%) for 48 hours	1. Simulate sensor humidity = 84% for 48 hours	Humidity = 84%, Duration = 48h	No outbreak alert is generated; status remains 'Normal'	As expected	Pass
TC-IOT-06	Temperature at lower boundary of risk band (20°C)	1. Simulate sensor temperature = 20°C with high humidity	Temperature = 20°C (lower bound)	Condition counted within the 20-25°C risk band for Late Blight evaluation	As expected	Pass

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-IOT-07	Temperature at upper boundary of risk band (25°C)	1. Simulate sensor temperature = 25°C with high humidity	Temperature = 25°C (upper bound)	Condition counted within the risk band; alert evaluated correctly	As expected	Pass
TC-IOT-08	Register a sensor using an invalid/duplicate sensor ID	1. Enter an already-registered sensor ID 2. Confirm	Duplicate sensor ID SM-1042	System rejects registration with 'Sensor already registered'	As expected	Pass
TC-IOT-09	Sensor reports an out-of-range impossible value	1. Simulate sensor sending temperature = 150°C	Invalid sensor payload, temperature = 150°C	Reading is discarded and flagged as a faulty-sensor anomaly, not used in risk analysis	As expected	Pass
TC-IOT-10	Sensor loses connectivity for an extended period	1. Disconnect sensor power for 24 hours 2. Observe dashboard	Sensor offline for 24 continuous hours	Dashboard marks sensor as 'Offline' and notifies the farm manager after 30 minutes of no data	As expected	Pass
TC-IOT-11	Weather API third-party service times out	1. Trigger weather-impact analysis while external API is unreachable	External weather API request timeout	System falls back to last cached weather data and logs the integration failure	As expected	Pass
TC-IOT-12	High-risk outbreak alert dispatch to all farmers in a grid zone	1. Trigger simulated high-risk condition 2. Verify notification dispatch	Grid zone with 25 registered farmers	All 25 farmers receive SMS and in-app alert with standardized wording within 1 minute	As expected	Pass
TC-YP-01	Generate yield forecast with valid crop, acreage and historical data	1. Open 'Yield Forecast' 2. Enter crop type, acreage, historical yield 3. Submit	Crop = Wheat, Acreage = 25 acres, 3 years history	System returns a predicted yield range with confidence interval	As expected	Pass
TC-YP-02	Compare forecast against previous season's actual yield	1. Generate forecast 2. Open comparison view	Current forecast + last season actuals	Comparison chart displays predicted vs. actual yield side by side	As expected	Pass
TC-YP-03	Enter minimum allowed acreage (0.1 acre)	1. Enter acreage = 0.1 2. Submit	Acreage = 0.1 (lower boundary)	System accepts the value and generates a scaled-down forecast	As expected	Pass
TC-YP-04	Enter maximum allowed acreage (10,000 acres)	1. Enter acreage = 10000 2. Submit	Acreage = 10000 (upper boundary)	System accepts the value and generates an enterprise-scale forecast	As expected	Pass

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-YP-05	Enter acreage just above maximum limit (10,001 acres)	1. Enter acreage = 10001 2. Submit	Acreage = 10001	System rejects with 'Acreage exceeds supported maximum of 10,000 acres'	As expected	Pass
TC-YP-06	Submit forecast request with negative acreage	1. Enter acreage = -5 2. Submit	Acreage = -5	System displays validation error 'Acreage must be a positive number'	As expected	Pass
TC-YP-07	Submit forecast with crop type left blank	1. Leave crop type field empty 2. Submit	Crop type = (blank)	System displays 'Please select a crop type' and blocks submission	As expected	Pass
TC-YP-08	Enter non-numeric text in acreage field	1. Enter 'abc' in acreage field 2. Submit	Acreage = 'abc'	System displays 'Acreage must be numeric' and blocks submission	As expected	Pass
TC-YP-09	Historical dataset for the region is unavailable	1. Select a newly onboarded region with no history 2. Request forecast	Region with zero historical records	System informs user that forecast confidence is limited and uses regional averages instead	As expected	Pass
TC-YP-10	Prediction model service crashes mid-request	1. Submit forecast while model service is restarting	Valid input, model service unavailable	Request queued and user notified 'Forecast will be ready shortly', auto-retried	As expected	Pass
TC-TR-01	Retrieve treatment recommendation after positive diagnosis	1. Complete a disease diagnosis (Powdery Mildew) 2. Open 'Recommended Treatment'	Diagnosis = Powdery Mildew	System displays organic remedy, chemical option with dosage, and safety precautions	As expected	Pass
TC-TR-02	View dosage guideline detail for chemical treatment	1. Open treatment card 2. Tap 'View Dosage Details'	Treatment = Copper Fungicide	Dosage instructions display '0.5% copper fungicide solution' with application frequency	As expected	Pass
TC-TR-03	Treatment recommendations are version-controlled and timestamped	1. Open a treatment card 2. Check metadata	Existing treatment record	Card shows last-updated timestamp and version number of the recommendation	As expected	Pass
TC-TR-04	Diagnosis confidence exactly at auto-recommendation threshold	1. Diagnosis returned at exactly 70% confidence 2. Check if treatment auto-displays	Confidence = 70% (threshold)	Treatment recommendation is auto-displayed (inclusive boundary)	As expected	Pass
TC-TR-05	Request treatment for a disease not present in the knowledge base	1. Force an unmapped disease code 2. Request treatment	Disease code = 'UNKNOWN_099'	System shows 'No verified treatment found, forwarding to agronomist review'	As expected	Pass

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-TR-06	Attempt to access treatment screen without completing diagnosis	1. Navigate directly to treatment screen via deep link 2. No prior diagnosis in session	No diagnosis ID in session	System redirects to diagnosis screen with message 'Please complete a diagnosis first'	As expected	Pass
TC-TR-07	Treatment knowledge base database connection fails	1. Request treatment while DB is unreachable	Valid diagnosis, DB connection error	System serves last cached recommendation if available, else shows graceful error message	As expected	Pass
TC-AN-01	Generate monthly PDF farm analytics report	1. Login as Farm Manager 2. Select month 3. Tap 'Export PDF'	Reporting month = July 2026, 40 farm records	PDF report generated with health trends, yield summary and disease incidence charts	As expected	Pass
TC-AN-02	Filter dashboard analytics by region and crop type	1. Open dashboard 2. Apply filters: Region = North, Crop = Rice	Filter = Region North, Crop Rice	Dashboard updates to show only matching farm records and aggregated metrics	As expected	Pass
TC-AN-03	Generate report for a farm manager with zero farms on record	1. Login with a new manager account 2. Request report	0 associated farm records	System displays 'No data available for this period' instead of an empty/broken report	As expected	Pass
TC-AN-04	Request report for a future month that has not occurred	1. Select month = December 2026 (future) 2. Export	Reporting month = future date	System blocks request with 'Cannot generate report for a future period'	As expected	Pass
TC-AN-05	Report export interrupted by session timeout	1. Begin PDF export 2. Session expires mid-generation	Session timeout = 30 minutes, export in progress	System resumes/re-authenticates and completes export without data loss	As expected	Pass
TC-SEC-01	Farmer logs in with valid credentials	1. Open app 2. Enter valid username and password 3. Tap Login	Valid registered account credentials	User is authenticated and redirected to the home dashboard	As expected	Pass
TC-SEC-02	Account lockout after repeated failed login attempts	1. Enter incorrect password 5 consecutive times	5 invalid login attempts	Account is temporarily locked for 15 minutes and user is notified via email	As expected	Pass
TC-SEC-03	Login attempt with invalid password	1. Enter valid username 2. Enter incorrect password 3. Submit	Valid username, invalid password	System displays 'Invalid username or password' without revealing which field is wrong	As expected	Pass
TC-SEC-04	Sensitive data is encrypted in transit	1. Capture network traffic during login and image upload	Live traffic during login/upload	All traffic uses TLS 1.2+ encryption; no plaintext credentials or images observed	As expected	Pass

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-NFR-01	Disease detection inference completes within SLA	1. Upload standard-size leaf image 2. Measure response time	Standard 2MB image, average network conditions	Response returned in under 3 seconds for 95% of requests	As expected	Pass
TC-NFR-02	Application usable on low-end Android device	1. Install app on a 2GB RAM Android device 2. Perform diagnosis flow	Entry-level Android smartphone	App loads and completes diagnosis without crashing or excessive lag	As expected	Pass
TC-NFR-03	Dashboard remains responsive under 1,000 concurrent users	1. Run load test with 1000 simulated concurrent dashboard users	1000 concurrent virtual users	Average page response time remains under 2 seconds with no failed requests	As expected	Pass
TC-NFR-04	System maintains 99.5% uptime over a monitoring period	1. Monitor system availability over 30 days	30-day uptime monitoring log	Recorded uptime meets or exceeds the 99.5% availability target	As expected	Pass

8. Requirement Traceability

Each functional and non-functional requirement is mapped below to the test cases from Section 7 that verify it, so that requirement coverage can be confirmed at a glance without a separate traceability table.

- FR1 / FR2 / FR3 (Disease detection, overlay, expert flagging) → TC-DD-01 to TC-DD-14
- FR4 / FR5 (IoT ingestion, outbreak risk alerts) → TC-IOT-01 to TC-IOT-12
- FR6 (Yield forecasting) → TC-YP-01 to TC-YP-10
- FR7 (Treatment recommendation) → TC-TR-01 to TC-TR-07
- FR8 (Weather impact analysis) → TC-IOT-11, TC-IOT-12
- FR9 (Analytics reports) → TC-AN-01 to TC-AN-05
- FR10 (Manager dashboard) → TC-AN-02, TC-NFR-03
- FR11 (Authentication and authorization) → TC-SEC-01 to TC-SEC-04
- NFR1 (Performance) → TC-NFR-01, TC-DD-01
- NFR2 (Scalability) → TC-DD-14, TC-NFR-03
- NFR3 (Reliability) → TC-NFR-04, TC-DD-13, TC-TR-07
- NFR4 (Security) → TC-SEC-02, TC-SEC-03, TC-SEC-04
- NFR5 (Usability) → TC-NFR-02
- NFR6 (Compatibility) → TC-AN-02
- NFR7 (Data Integrity) → TC-IOT-09

9. Test Coverage Summary

A total of 47 test cases were designed across five functional modules and one cross-cutting security/non-functional group. Of these, 14 test cases target the Disease Detection module, 12 target IoT and Weather Monitoring, 10 target Yield Prediction, 7 target Treatment Recommendation, 5 target Analytics and Reporting, and the remaining 8 cover authentication, security, and non-functional quality attributes.

Coverage by condition type is balanced across the suite: approximately 19 test cases exercise normal (positive) paths, 11 apply boundary value analysis at threshold conditions, 10 validate handling of invalid input, and 11 exercise exceptional or failure conditions such as network loss, service outages, and data anomalies. Every functional requirement listed in Section 3 is covered by at least one test case, and every non-functional requirement in Section 4 is covered by at least one measurable test case, giving the suite complete requirement coverage at the level of this assignment.

All eleven functional requirements (FR1–FR11) and all seven non-functional requirements (NFR1–NFR7) identified in Sections 3 and 4 have at least one associated test case in the traceability mapping in Section 8, with no orphaned requirements and no test case left unmapped to a requirement.

10. Conclusion

This document has identified the functional and non-functional requirements of the Intelligent AI-Based Smart Crop Disease Diagnosis and Yield Prediction System, mapped those requirements onto the system's layered architecture, defined representative test scenarios for every module, and applied Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, State Transition Testing, and Error Guessing to design a complete set of 47 test cases. The accompanying traceability mapping and coverage summary demonstrate that the resulting test suite provides adequate requirement and functional coverage for the assigned problem statement, fulfilling the deliverable requested in the assignment instructions.